EDA Case Study: Google Play Store

Workshop 4 — Module 7

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Data Visualization for Data Scientists
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Module Roadmap

1. Data cleaning audit trail: 10,841 $\rightarrow$ 9,360 rows
2. Apply the four-phase EDA workflow to a real dataset
3. Build a nine-panel EDA dashboard
4. Extract and communicate four key findings
5. Outlier deep-dive by category

Learning Outcome

You will execute a complete EDA pipeline on the Google Play Store dataset, integrating every technique from W04-M01 through M06 (cleaning, missingness, outliers, correlation) into a single coherent analysis.

Phase 1: Clean & Document

Data Cleaning Audit Trail

Data Cleaning Audit Trail: 10,841 → 9,360 rows

Step	Rows	Change
Raw rows	**10,841**	—
After dedup	**9,660**	Removed 1,181 duplicate App names
After type fix	**9,660**	Reviews, Installs, Price → numeric
After NA drop	**9,367**	Dropped 293 missing Rating
After type filter	**9,367**	Kept Free + Paid only
After range filter	**9,360**	Removed 7 invalid Rating (< 1 or > 5)

Pro-Tip

Every cleaning step must be **documented and justified**. An audit trail (raw → dedup → type-fix → NA drop → filter) makes your analysis reproducible and defensible. Include row counts at each step.

The Cleaning Pipeline (R)

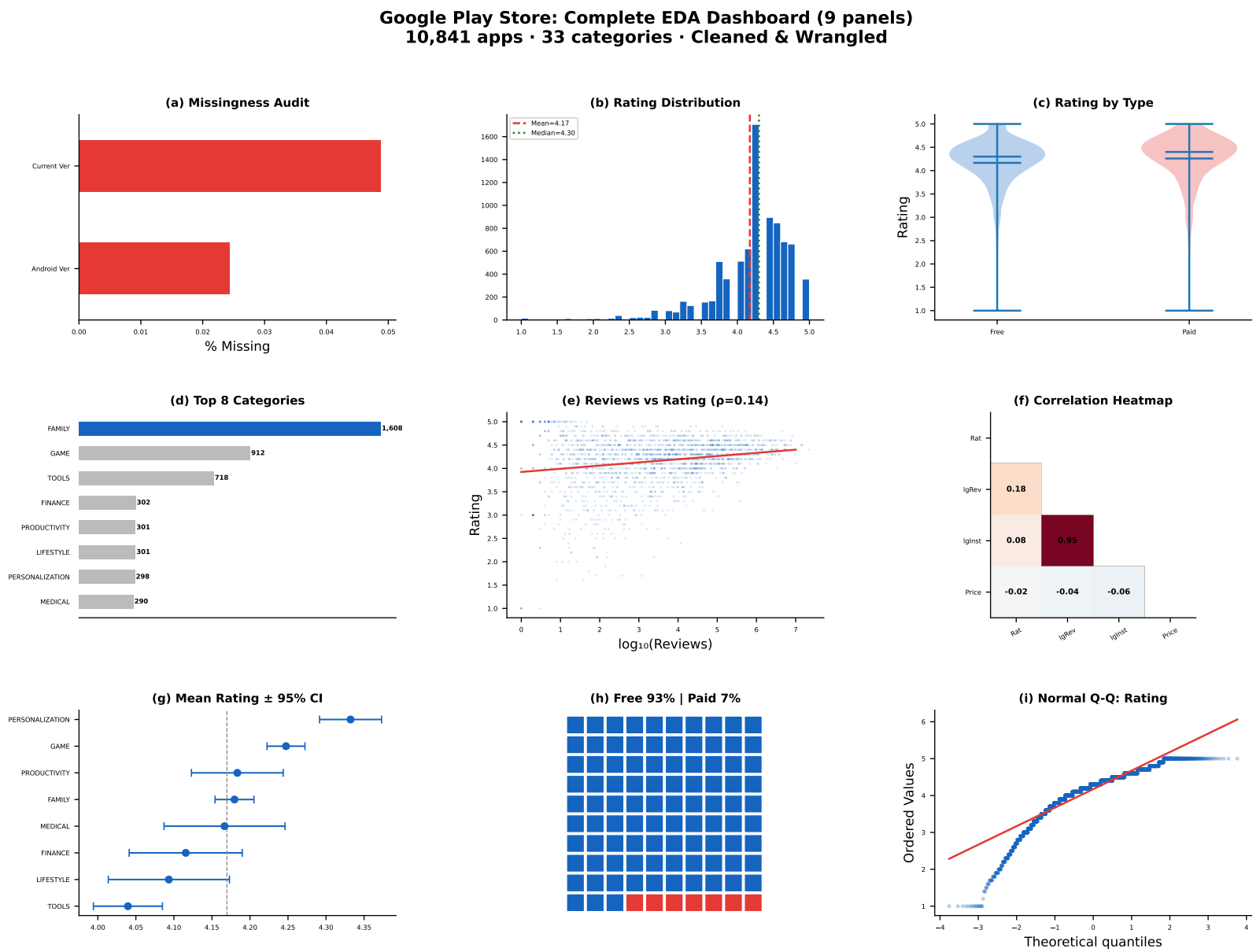
```r
apps_clean <- read_csv("googleplaystore.csv") |>
  # Step 1: Remove 1,181 duplicate app entries
  distinct(App, .keep_all = TRUE) |>
  # Step 2: Fix column types
  mutate(
    Reviews  = as.numeric(Reviews),
    Installs = str_remove_all(Installs, "[+,]") |> as.numeric(),
    Price    = str_remove(Price, "\\$") |> as.numeric(),
    log_reviews = log10(pmax(Reviews, 1)),
    log_installs = log10(pmax(Installs, 1))
  ) |>
  # Step 3: Drop missing Rating (293 rows)
  drop_na(Rating) |>
  # Step 4: Keep valid Type
  filter(Type %in% c("Free", "Paid"), Rating >= 1, Rating <= 5) |>
  # Step 5: Factor ordering for ggplot
  mutate(Category = fct_lump_n(Category, 8) |> fct_infreq())
# Result: 9,360 rows x 15 columns (audit: 10841 -> 9660 -> 9367 ->
    9360)
```

The Cleaning Pipeline (Python equivalent)

```
apps_clean = (
    pd.read_csv("googleplaystore.csv")
    .drop_duplicates(subset="App", keep="first")          # 10841 ->
          9660
    .assign(
        Reviews=lambda d: pd.to_numeric(d["Reviews"], errors="coerce
            "),
        Installs=lambda d: d["Installs"].str.replace(r"[+,]", "",
            regex=True)
            .pipe(pd.to_numeric, errors="coerce"),
        Price=lambda d: d["Price"].str.replace("$", "", regex=False)
            .pipe(pd.to_numeric, errors="coerce"))
    .dropna(subset=["Rating"])                              # -> 9367
    .query("Type in ['Free','Paid'] and Rating>=1 and Rating<=5")  #
          -> 9360
    .assign(
        log_reviews=lambda d: np.log10(d["Reviews"].clip(lower=1)),
        log_installs=lambda d: np.log10(d["Installs"].clip(lower=1))
            )
)
```

Phase 2: The Nine-Panel Dashboard

Complete EDA Dashboard



Source: Kaggle Google Play Store | UNYT Data Visualization Course

Reading the Dashboard: Panel by Panel

#	Panel	EDA Question	Technique (Module)
(a)	Missing % bar	How much is missing?	M04 (naniar/missingno)
(b)	Histogram + lines	What shape is Rating?	M01 (Tukey 4-plot)
(c)	Violin by Type	Do Free/Paid differ?	W03-M02 (violin)
(d)	Sorted bar	Which categories dominate?	W03-M03 (bar)
(e)	Scatter + trend	Reviews $\rightarrow$ Rating?	M06 (correlation)
(f)	Corr. heatmap	Which vars correlate?	M06 (heatmap)
(g)	Pointrange	Group means differ?	W03-M07 (CI)
(h)	Waffle	Free/Paid split?	W03-M05 (proportion)
(i)	Q-Q plot	Is Rating normal?	M01 (diagnostics)

Building the Dashboard (R)

```r
library(patchwork)
# (a) Missing audit    (b) Histogram        (c) Violin
# (d) Sorted bar       (e) Scatter           (f) Heatmap
# (g) Pointrange       (h) Proportion        (i) Q-Q

# Compose: 3 x 3 grid
dashboard <- (p_miss | p_hist | p_violin) /
             (p_bar  | p_scatter | p_heat) /
             (p_point | p_waffle | p_qq) +
  plot_annotation(
    title = "Google Play Store: Complete EDA Dashboard",
    subtitle = "9,360 apps | 33 categories | Cleaned & Wrangled",
    caption = "Source: Kaggle | UNYT Data Viz Course",
    tag_levels = "a") &
  theme_minimal(base_size = 8)

ggsave("eda_dashboard.pdf", dashboard, width = 16, height = 11)
```

Phase 3: Extract Findings

Four Key Findings

EDA Key Findings: Four Insights from the Dashboard

#	Finding	Evidence	Implication
1	**Rating distribution is left-skewed (ceiling at 5.0)**	Most apps cluster at 4.0–4.5; few below 3.0	**Median > Mean confirms skew**
2	**Free apps dominate (93% Free, 7% Paid)**	Waffle chart shows 93 of 100 squares are blue	**Imbalanced groups: stratify any comparison**
3	**Reviews correlate weakly with Rating (ρ = 0.18)**	Scatter is a cloud, not a line; high-review apps span full range	**Log transform essential (4 orders of magnitude)**
4	**MEDICAL has lowest mean rating; SOCIAL has highest**	Pointrange CIs overlap for most categories	**Differences may not be statistically significant**

Finding 1: Left-Skewed Ratings (Ceiling Effect)

- Mean Rating = 4.17, Median = 4.30 (median > mean ⇒ left skew)
- Most apps cluster at 4.0–4.5; very few below 3.0
- Q-Q plot confirms departure from normality at both tails
- **Implication**: use **Spearman** (not Pearson) for correlations; **median** (not mean) for central tendency; consider **non-parametric** tests

Finding 2: Massive Free/Paid Imbalance (93/7%)

- 93% of apps are Free; only 7% Paid
- Any comparison between Free and Paid must account for this imbalance: confidence intervals for Paid are much wider
- **Implication**: stratify by Type before comparing categories; use **bootstrap** or **weighted** methods to avoid sample-size artefacts

Finding 3: Reviews $\leftrightarrow$ Rating Is Weak ($\rho = 0.18$)

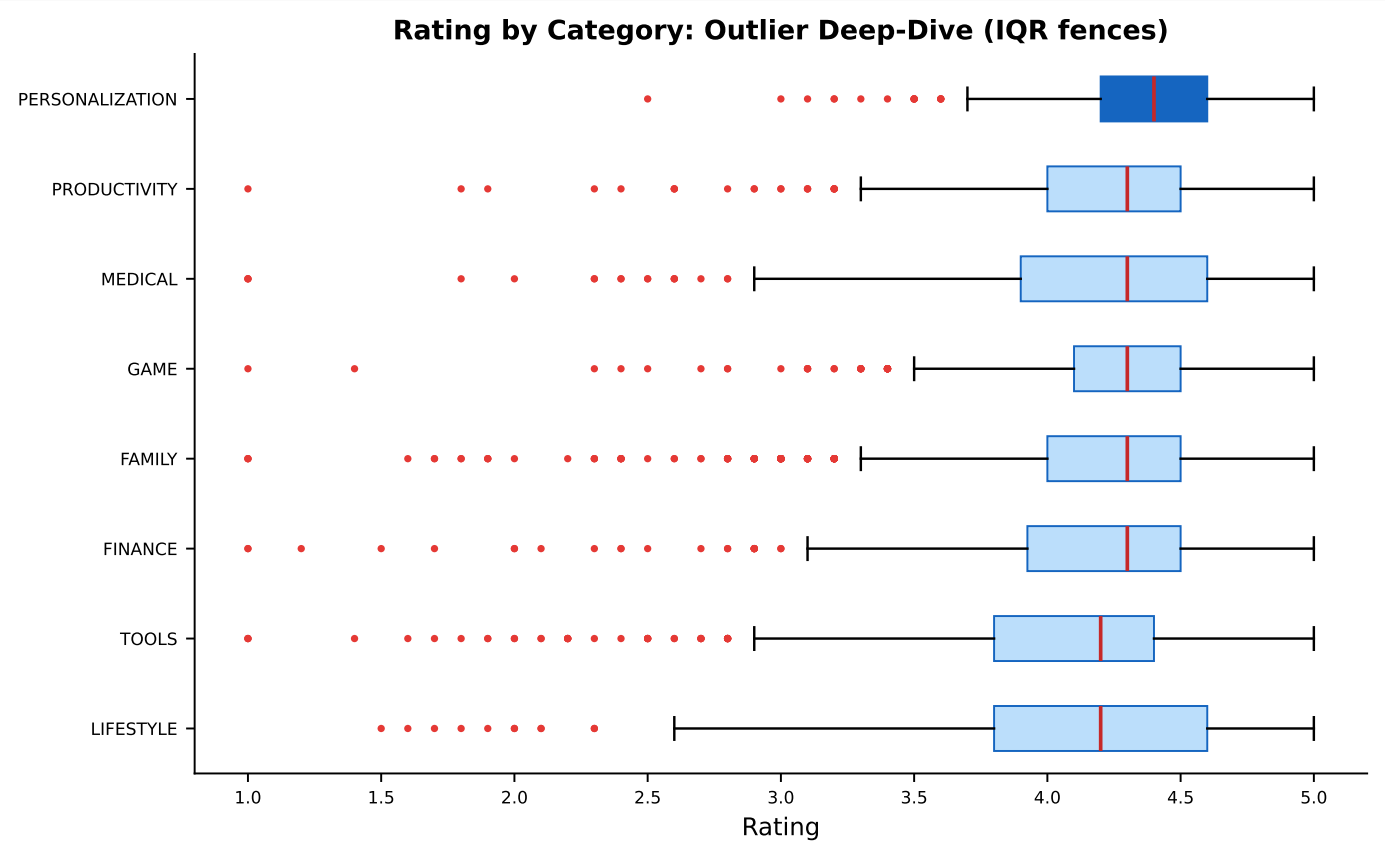
- Spearman $\rho = 0.18$: statistically significant (large n) but practically negligible
- The scatter is a cloud, not a line: high-review apps span the full 1–5 rating range
- Log transform of Reviews is essential (range: 0 to 78M)
- **Implication**: popularity (reviews/installs) does not predict quality (rating). Don't use one as a proxy for the other.

Finding 4: Category Differences Are Small

- SOCIAL (4.30) has the highest mean; MEDICAL (3.95) the lowest
- But 95% CIs overlap for most categories $\Rightarrow$ differences are likely not significant after multiple testing
- **Implication**: category is a weak predictor of rating; within-category variance dominates between-category variance

Phase 4: Outlier Deep-Dive

Outlier Deep-Dive by Category



Data Insight

All categories show left-tail outliers (low-rated apps). GAME and FAMILY have the most outliers by count (largest n). The outliers are not data errors — they represent genuinely poorly-rated apps. Decision: keep them, as they represent the real distribution.

Synthesis

The EDA Report Structure

Every EDA case study follows this structure:

Section	Content
1. Data Description	Source, n, variables, types
2. Cleaning Audit	Steps, row counts, justifications
3. Missingness Audit	Matrix, %, mechanism hypothesis
4. Univariate EDA	Distributions, outliers, normality
5. Bivariate EDA	Scatter, correlation, grouped comparisons
6. Key Findings	3–5 numbered insights with evidence
7. Limitations	What you can't conclude, next steps

This structure applies to **every** dataset you will encounter in this course (Netflix, e-Car, CRM, Real Estate) and in professional practice.

Key Takeaways

1. **Audit trail**: document every cleaning step with row counts (10,841 $\rightarrow$ 9,360).
2. **Dashboard first**: a 6–9 panel figure gives a helicopter view before diving into individual charts.
3. **Findings $\neq$ plots**. Extract numbered insights with three columns: finding, evidence, implication.
4. Google Play Store: ratings are ceiling-skewed, Free dominates 93%, Reviews $\not\rightarrow$ Rating, category differences are small.
5. The **EDA report structure** (7 sections) applies to every dataset.

Essential Readings

- Wickham, H. & Grolemund, G. (2023). *R for Data Science*, 2nd ed., Chapter 10 (Exploratory Data Analysis).
- Peng, R. (2016). *Exploratory Data Analysis with R*. Leanpub. (Free online.)
- Google Play Store Dataset: Kaggle. `https://www.kaggle.com/lava18/google-play-store-apps`

Next Module: EDA Case Study — e-Car Loan Pricing (W04-M08)